Frames and matrix designs

Vanishing injectivity probability at 4d minus 5 complex phase measurements

Difficulty: challenging **Importance:** interesting to the community

Status: Lean verified

Rating rationale: The asymptotic typical-injectivity question needs quantitative algebraic and probabilistic control beyond positive failure probability; it informs phase-retrieval measurement design.

Original statement

For every integer $d \geq 2$, draw $A_d \in \mathbb{C}^{(4d-5)\times d}$ with independent standard complex Gaussian entries: the real and imaginary parts are independent $N(0, 1/2)$. Let p_d be the probability that

$$|A_d x| = |A_d y| \quad \Longrightarrow \quad y = e^{\mathrm{i}\theta} x \text{ for some } \theta \in \mathbb{R}$$

holds for all $x, y \in \mathbb{C}^d$, where absolute value is componentwise. Equivalently, p_d is the probability that the induced phase-retrieval map on vectors modulo global phase is injective.

Is

$$\lim_{d\to\infty} p_d = 0?$$

This is part (b) of Vinzant's conjecture as restated in Randomstrasse 101. It distinguishes injective exceptional measurement systems from typical matrices at a row count just below $4d-4$. Injectivity is the basic identifiability requirement before conditioning and stable numerical inversion can be addressed.

Resolution — independently audited 8 October 2026

Zhangsong Li's 13-page manuscript, Theorem 1.4, proves the stronger bound

$$p_d \leq C/d \qquad (d \geq 2)$$

for an absolute constant $C > 0$. This implies the displayed limit. The argument plants a two-dimensional ambiguity, compares its law in L^2 to a covariance-matched Gaussian law, and uses a quantitative implicit-function argument to turn the planted approximate ambiguity into an exact one.

An independent AI-agent proof audit checked the manuscript's planted and reference laws, the cone and overlap estimates, the small-ball and Newton arguments, and the final probability reduction. No proof gap was found. That informal audit is separate from the Lean verification recorded below; neither is external human peer review. The original target, permanent identifier, and canonical path are unchanged.

Local Lean development — 27 September 2026

The Lean solution proves the bound $p_d \leq C/d$ for every $d \geq 2$ and the original limit $p_d \to 0$, using the original Gaussian law and all-signals injectivity predicate. See the final assembly, source guide, and local verification record. The subsequent independent review and isolated Linux verification are in the evidence section.

Lean proof and verification evidence

Reviewed 13 October 2026.

At immutable repository revision *bca7904b*, *NLA.FR05.phaseRetrieval_injective_probability_le_inv* proves one positive absolute constant C bounds the original injectivity probability by C/d for every $d \geq 2$; *NLA.FR05.phaseRetrieval_injective_probability_tendsto_zero* proves the retained limit. The independent mathematical and statement review checks the original iid complex-Gaussian law at $4d - 5$ measurements, the all-signals quantifier, the global-phase equivalence, and the asymptotic conclusion. No extra mathematical premise is used.

The project pins Lean **4.33.1**, Mathlib *0df444a360ea*, and LeanCert *621a43d7cf21* in its toolchain and manifest. From *frames-and-matrix-designs/FR-05/lean*, run *lake build Challenge Solution*; the 9 October isolated Linux Comparator/LeanCert job and log also checks the selected exports against the challenge and passes kernel trust. Its transitive axiom reports contain only *propext*, *Classical.choice*, and *Quot.sound*, with no proof admission or custom axiom. The receipt records source hashes and reproduction details. This catalog reviewed that public verification record; it did not rerun Lean locally for this status update.

References

1. A. S. Bandeira et al., *Randomstrasse 101: Open Problems of 2025*, arXiv:2603.29571 (2026), Conjecture 19(b). Paper.
2. C. Vinzant, *A small frame and a certificate of its injectivity*, SAMPTA 2015, arXiv:1502.04656. The explicit 11-vector frame in $\mathbb{C}^4$ demonstrates why the below-$4d-4$ regime cannot simply be dismissed. Paper.
3. Z. Li, *Resolution of Vinzant's Conjecture on Phase Retrieval Injectivity* (13 September 2026), Theorem 1.4. The manuscript proves $p_d \leq C/d$ for every $d \geq 2$; the audited copy is pinned by SHA-256 in the review. Manuscript.
4. Z. Li, *On Injectivity of Phase Retrieval*, arXiv:2606.17922 (2026). This earlier version established only $p_d < 1$. Paper.

Historical status check — 2026-09-10

Searched "Vinzant conjecture injectivity probability limit", "phase retrieval 4M-5 2026", and checked the latest June 2026 paper abstract/version. Its positive probability of noninjectivity does not supply an asymptotic lower bound tending to one. The already proved assertion $p_d < 1$ is deliberately excluded from this problem.

Audit update (2026-09-10): Rechecked Li's June 2026 abstract and searched for the asymptotic part of Vinzant's conjecture. The reported strict inequality $p_d < 1$ does not force $p_d \to 0$; this entry contained only the latter target. This was a bounded literature check, not a proof that no solution existed at that time.